

Network Security

Module 02

Duration: 24:29

Video: <https://www.youtube.com/watch?v=kibMPvGrLXw>

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hello everyone I am pradep bandari from duaba college jander I'm going to present a session on network security over the past few years the internet enabled business or E business has drastically improved company's efficiency and revenue growth e business applications such as e-commerce Supply Chain management and remote assess enable companies to streamline processes lower operating costs and increase customer satisfaction such applications require Mission critical networks that accommodate voice video and data traffic and these networks must be scalable to support increasing number of users and the need for greater capacity and performance however as networks enable more and more applications and are available to more and more users they

become ever more vulnerable to a wider range of security threats to combat these threats and ensure that e business transactions are not compromised network security play a major role in today's networks network security deals with securing the information through cryptography and network security devices the main goal of net network security confidentiality now confidentiality is probably the most common aspect of information security we need to protect our confidential information an organization needs to guard against those malicious actions that endanger the confidentiality of its information confidentiality not only applies to the storage of the information it also applies to the transmission of information

when we send a piece of information to be stored in a remote computer or when we retrieve a piece of information from a remote computer we need to conceal it during transmission second is integrity information needs to be changed constantly in a bank when a customer deposits or withdraws money the balance of her account needs to be changed in integrity means that changes need to be done only by authorized entities and through authorized mechanisms Integrity violation is not necessarily the result of a malicious act and interruption in the system such as a power failure or a power surge may also create unwanted changes in some information third is availability the third component of information security

the information created and stored by an organization needs to be available to authorized entities information is useless if it is not available information needs to be constantly changed which means it must be accessible to authorized entities the unavailability of information is

just as harmful for an organization as the lack of confidentiality or Integrity imagine what would happen to a bank if the customers could not access their accounts for the transaction security attacks now any action that compromises the security of information owned by an organization is called security attack two things to be considered separately in context of security attacks are threat a potential for violation of security which exists

when there is a circumstance capability action or event that could breach security and cause harm that is a threat is a possible danger that might exploit a vulnerability second is attack an assault on system security that derives from an intelligent threat that is an intelligent act that is a deliberate attempt to evade Security Services and violate the security policy of a system now security attack attacks can be of two types these are passive attacks and active attacks passive attacks passive attacks are in nature of if dropping on or monitoring of transmission the goal of the opponent is to obtain information that is being transmitted two types of passive attacks are release of message

contents and traffic analysis release of message contents is easily understood a telephone conversation an electronic message and a transferred file may contain sensitive or confidential information we would like to prevent an opponent from learning the contents of these Transmissions the figure on the screen reveals the concept of if dropping a second type of passive attack is traffic analysis suppose that we have had a way of masking the contents of messages or other information traffic so that the opponents even if they captured the message could not extract the information from the message the common technique for masking contents is encryption if we had encryption protection in place an opponent might still be able to observe the pattern of these messages the opponent could

determine the location and identity of communicating hosts and could observe the frequency and length of messages being exchanged the information might be useful in guessing the nature of the communication that was taking place the figure on the screen reveals the concept of traffic analysis replay involves the passive capture of a data unit and its subsequent retransmission to produce an unauthorized effect as shown on the screen modification of messages simply means that some portion of a legitimate message is altered or that messages are delayed or reordered to produce an unauthorized effect for example a message meaning allow John Smith to read confidential file accounts is modified to mean allow Fred Brown to read confidential file

7 minutes, 1 second counts the concept is shown on the screen the denial of service prevents or inhibits the normal use or management of communication facilities this attack may have a specific Target for example an entity May suppress all messages directed to a particular destination another form of service denial is the disruption of an entire network either by disabling the network or by overloading it with messages so as to degrade performance the concept is shown on the screen are three goals of security that is confidentiality integrity and availability can be threatened by security attacks we can divide security attacks into three

groups related to the security goals as well. A New Concept snooping is introduced in this

classification snooping means unauthorized access to or interception of data. For example, a file transferred through the internet may contain confidential information. An unauthorized entity may intercept the transmission and use the content for her own benefit. To prevent grouping the data can be made meaningless for the interceptor by using encryption techniques. Let us discuss various network security service techniques. Some security services are required to achieve security goals and prevent attacks. The actual implementation of security goals needs two techniques. These techniques are designed to protect one or more attacks while maintaining security goals. The first technique is cryptography and second is

security through specific devices. Cryptography, a word with Greek origin, means secret writing. However, we use the term to refer to the science and art of transforming messages to make them secure and immune to attacks. Although in the past cryptography referred only to the encryption and decryption of messages using secret keys, today it is defined as involving three distinct mechanisms: these are symmetric key cryptography, asymmetric key cryptography, and hashing. Let us discuss symmetric key cryptography. It is also known as traditional Cipher. In symmetric key cryptography, the same key is used for encryption and decryption, and the key can be used for bidirectional communication. The figure on the screen represents the general concept of

symmetric key cryptography. In this model, plain text is used as input. We convert the plain text into Cipher text by using an encryption algorithm and key. On the receiver side, Cipher text is converted into plain text by using decryption algorithm and key. In symmetric cryptography, sender and receiver share the same secret key. Asymmetric algorithms rely on one key for encryption and a different but related key for decryption. These algorithms have the following important characteristic: it is computationally infeasible to determine the decryption key given only knowledge of the cryptographic algorithm and the encryption key. In addition, some algorithms such as RSA also exhibit the following characteristics: either of the two related keys can be used for encryption with the other used for decryption. The

main components of asymmetric key cryptography are as shown on the screen. Plain text: this is the readable message or data that is fed into the algorithm as input. Encryption algorithm: performs various transformations on the plain text. Public and private keys: this is a pair of keys that have been selected so that if one is used as encryption key, the other is used as decryption key. The exact transformations performed by the algorithm depend on the public or private key that is provided as input. Cipher text: this is the scrambled message produced as output. It depends on the plain text and the key. For a given message, two different keys will produce two different Cipher texts. Decryption algorithm: this algorithm accepts the

cipher for text and the matching key and produces the original plain text hash functions a cryptographic hash function takes a message of arbitrary length and creates a message digest of fixed length all cryptographic hash functions need to create a fixed size digest out of a variable size message creating such a function is best accomplished using iteration instead of using a hash function with variable size input a function with fixed size input is created and is used as necessary number of times the fixed size input function is referred to as a compression function it compresses an n bit string to create an M bit string where n is normally greater than M the scheme is referred to as has an iterated cryptographic hash function several hash algorithms were designed by Ron rivest these are

referred to as md2 md4 md5 where MD stands for message digest the last version md5 is a strengthened version of md4 that divides the message into blocks of 512 bits and creates a 128bit Digest it turns out that a message digest of size 128 bits is too small to resist attack the secure hash algorithm sha is a standard that was developed by the National Institute of Standards and Technology called nist sha has gone through several versions now let's discuss the network security through specific network devices when we think about network security we can also 14 minutes, 1 second think about the boundary in which our network is established we need to protect our network from external attacks and we also need to mind internal suspicious activities we usually deploy security devices as shown

in the screen to protect our Network border routers as static filtering device firewalls in in detection systems idss IPS that is intrusion prevention systems border routers as static filtering devices now routers are the traffic cops of the network they direct traffic into out of and within our networks as you can observe in the figure the Border router is the last router you control before an untrusted Network such as the internet because all of internet traffic goes through this router it often functions as in Network's first and last line of defense through initial and final filtering we can configure border router as static packet filtering device by configuring ACL that is access control lists we called router as static filtering device because it can inspect

packets only up to layer three that is Network layer improperly destined traffic might be internal addresses that hit your external interface or vice versa and they can be addressed with Ingress and egress filtering border routers can also block traffic that is considered high risk from entering your network icmp is a favorite of attackers both for denial of service attack and Reon insense so blocking this protocol in whole or in part is a common function of a border router you may also consider blocking Source rooted packets at the border router because they can circumvent defenses the Border router can also Block outof band packets such as syin packets on February 9 2000 websites such as Yahoo and CNN were temporarily taken

off the internet mostly by distributed denial of service Smurf attacks a Smurf attack involves sending spoofed icmp Echo request that is pings to the broadcast addresses resulting in a response from every host in this case the spoofing allowed attackers to direct the large number of responses to a victim Network Ingress and egress filtering would have blocked the spoofed traffic and allowed them to weather the distributed denial of service storm every Network should have Ingress and Egress filtering at the border router to permit only traffic that is destined for the internal Network to enter and traffic that is destined for the external network to exit Firewall a firewall is a choke point device that has a set of rules specifying what traffic it will allow or deny to pass through it a firewall

typically picks up where the Border router leaves off and makes a much more thorough pass at filtering traffic firewalls can be of three types static stateful and proxy Firewalls although firewalls aren't perfect effect they do block what we tell them to block and allow what we tell them to allow as previously discussed router can be used as static packet filter or as static firewall but we need specialized firewalls to implement as stateful and proxy firewalls unlike static packet filtering devices stateful firewalls keep track of Connections in a state table and are the most common type of firewalls a stateful firewall blocks traffic that is not in its table of established connections the firewall rule base determines the source and destination IP and port numbers

permitted to establish connections by rejecting non-established non-permitted connections a stateful firewall helps to block reconnaissance packets as well as those that may gain more extensive unauthorized access to protected resources proxy firewalls are the most advanced and least common type of firewalls proxy firewalls are also stateful in that they block any non-established non-permitted connections as with stateful firewalls the firewall rule base determines the source and destination IP and port numbers that are permitted to establish connections proxy firewalls offer a high level of security because internal and external hosts never communicate directly rather the firewall acts as an

intermediary between hosts proxy firewalls examine the entire packet to ensure compliance with the protocol that is indicated by the destination port number ensuring that only protocol compliant traffic passes through the firewall helps defense in depth by diminishing the possibility of malicious traffic entering or exiting from your network proxy firewalls diminishes the possibility of malicious traffic entering or exiting your network by ensuring that only protocol compliant traffic passes through if intrusion detection systems now an IDS is like a security alarm system for your network that is used to detect an alert on malicious events the system might comprise many different IDS sensors placed at strategic points in your network in

general IDS sensors watch for predefined signatures of malicious events and they might perform statistical and anomaly analysis when IDS sensors detect suspicious events they can Alert in several different ways including email paging or simply logging the occurrence a

network IDs could identify and alert on the following first is DNS Zone transfer request from unauthorized host uni code attacks directed at a web server buffer overflow attack V propagation Etc intrusion prevention systems now an IPS is a system that automatically detects and prevents Network and hosts attacks in contrast to a traditional IDs which focuses on notifying the administrator of anomalies and IPS strives to automatically defend the target without administrators direct

involvement protection may involve using signature-based or behavioral techniques to identify an attack and then blocking the malicious traffic or system call before it causes harm in this respect an IBS combines the functionality of a firewall and IDs to offer a solution that automatically blocks offending actions as soon as it detects and attack frequently we get caught up in the technical aspect of network security without considering its non-technical elements tasks such as optimizing the firewall rule base examining Network traffic for suspicious patterns and locking down the configuration of systems are certainly important to network security what we often forget is the human end of things such as has policies and awareness that go along with the Technical Solutions policy determines what

security measures your organization should Implement as a result the security policy guides your decisions when implementing security of the network an effective defense and depth infrastructure requires a comprehensive and realistic security policy Hallmarks of good security policy include Authority who is responsible scope who it affects expiration when it ends specificity what is required and Clarity can everyone understand it so in this session we have discussed what is network security security issues and basic network security techniques I hope you have enjoyed the session thank you so much much 24 minutes, 1 second [Music]